KneeScope project plan

Knee MRI research review prototype

Learn2Earn group project | 13 October 2026 | Working product name

Recommended outcome: build a focused knee MRI research review tool with a real model, a usable viewer, and honest evaluation. The project is a standalone hackathon build. It is not a Kaggle submission and this plan contains no schedule.

This document brings together the complete product and engineering plan, the architecture diagram, the interface concept, ownership, verification criteria, and sources. The team should use one agreed scope rather than implement the competing requirements across the original PRD.

PRD interpretation: Section 17 of the supplied PRD supersedes the earlier competition requirements. It drops the Kaggle submission and narrows the intended model to four findings. The earlier sections remain useful background, but do not define additional must-have work. [1]

Execution principle: the concern that this project will waste time is an assumption to test, not a foregone conclusion. Uncontrolled scope and unclear ownership are the larger risks. Each person should have a bounded, reviewable contribution.

1 Define the product clearly

Product statement: KneeScope lets a researcher inspect a knee MRI study, review model scores for supported abnormalities, and examine the model's limitations.

The primary user for this build is a researcher or learner reviewing MRI studies. Radiologists are the eventual audience, but the prototype has not established a benefit to their clinical workflow.

1. Select an available study.
2. Inspect its imaging series.
3. Run inference.
4. Review scores for supported findings.
5. Inspect influential slices, if explainability is implemented.
6. Compare predictions with dataset labels in a separate evaluation view.
7. See model version, data source, limitations, and measured performance.

2 Let the dataset determine the findings

The RSNA competition covers twelve abnormalities and evaluates macro-averaged ROC AUC. This project can intentionally select a smaller subset. [2]

MRNet is not a drop-in replacement for the proposed four outputs. Its published tasks are overall abnormality, ACL tear, and meniscal tear. It does not provide the required MCL and separate medial and lateral meniscus targets. [3]

Available data	Supported product scope
RSNA with suitable labels and permitted hackathon use	ACL, MCL, medial meniscus, lateral meniscus
MRNet with approved access and suitable terms	Overall abnormality, ACL tear, meniscal tear
Organizer-provided dataset	Exactly the findings its labels support
No usable training data	Clearly identified interaction prototype; no detection claims

Prefer accessible data with reliable labels over preserving four attractive result cards. Before committing, record access and permitted training, hosting and demonstration uses; file format and size; available series; whether labels are supplied or extracted; positive, negative and unknown label counts; and whether patient and site identifiers support grouped evaluation.

A local demo or recorded video does not automatically resolve data-use restrictions. Those permissions still need checking. The exact dataset and permissions remain open decisions.

3 Freeze a small complete scope

Must deliver	Add after the core works	Leave out
Dataset audit and supported finding list	Influential-slice navigation	Kaggle submission pipeline
Shared preprocessing	Grad-CAM overlays	Report and image fusion
One trained baseline	Calibration if sample size supports it	Ensembles and extensive tuning
Honest held-out evaluation	Exportable research summary	Accounts, billing and EHR integration
Study picker and series viewer	General upload support	Diagnostic or treatment recommendations
Model scores and failure states	Additional supported findings	Segmentation or injury grading
Versioned live and cached results	Reviewer notes	Clinical deployment
README and model card		

A metrics page belongs in the core scope. It can be simple, but the audience needs evidence that the model was evaluated.

4 Design three connected screens

Study browser

Show study identifier, available planes, series count and analysis status. Display only metadata needed for the task.

Review workspace

Use a dark charcoal workspace with restrained teal accents, a large imaging area and a compact findings panel.

- Sagittal, coronal and axial series tabs where available.
- Slice scrolling, slider, zoom, reset and keyboard controls.
- Clear missing-series states.
- Run analysis action and visible processing status.
- Findings with model scores and support status.
- Model version and a live or cached result badge.
- Persistent research-use notice.

Keep unsupported, not evaluated and failed distinct from low scores.

Evaluation and model card

Show dataset, split method, sample counts, per-finding metrics, inference measurements and known failure cases. Ground-truth labels appear here or behind an explicit Reveal dataset label action. They must never feed inference.

Viewer corrections: three independently acquired planes are not necessarily synchronized views of the same location. Do not imply synchronization. Brightness and contrast changes on rendered PNGs are display adjustments; true MRI window and level require underlying intensity data or server-side rendering.

The concept on the following page contains illustrative imaging and no real predictions. Its four finding cards reflect the proposed RSNA scope; the final UI must adapt to the selected dataset.

Interface concept



Figure 1. KneeScope review workspace concept. All imagery, metadata and attention overlays are illustrative. This is a design reference, not medical evidence or an implemented product.

The large study viewer remains the visual focus. Findings sit beside it, with explicit support status. Evaluation is a separate destination, and the research prototype notice remains visible. The mockup's metadata and series labels are illustrative and must not be copied as real study information.

5 Use one straightforward model pipeline

Start with a pretrained 2D backbone processing slices, then pool features into a study-level prediction. Adjacent-slice stacks can be an experiment rather than the first implementation.

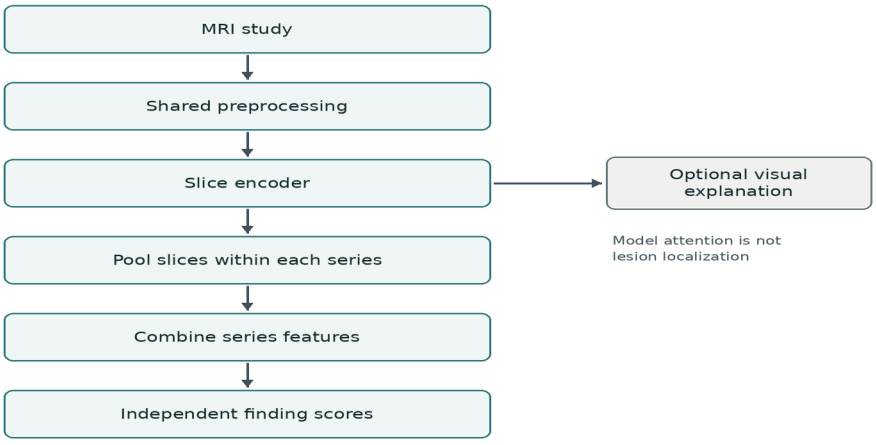


Figure 2. Proposed model pipeline. The optional visual explanation branches from the slice encoder. It does not establish the location of a lesion.

- One manageable backbone and a documented slice sampling strategy.
- Fixed input dimensions and intensity normalization.
- Explicit handling of missing planes.
- Independent sigmoid outputs for supported findings.
- Binary cross-entropy; introduce weighting only when justified by label imbalance.
- Unknown labels masked from loss and evaluation.
- Saved configuration, seed, checkpoint and preprocessing version.

Begin with a frozen encoder and a trained prediction head. Fine-tune if the baseline and compute budget justify it. Do not blindly mirror knees or sort DICOM files by filename. Preserve orientation information, use spatial metadata where available, and visually inspect preprocessing output.

If labels come from reports, use controlled extraction that handles negation and uncertainty, preserves provenance, and receives manual spot checks. The original report must not become an inference shortcut for predicting labels extracted from that same report.

6 Evaluate without misleading yourselves

The evaluation unit is the study, with patient grouping wherever available. Slices from the same study must never cross splits. Separate training data, validation data used for choices and thresholds, and a final untouched test partition if the dataset permits one.

If only a validation set is available, call the result validation performance. Do not rename it a test result after repeated tuning.

Measurement	**Purpose**
Per-finding ROC AUC	Ranking performance
Macro AUC	Summary across evaluable findings
PR AUC and prevalence	Interpret performance on imbalanced labels
Sensitivity and specificity at a declared threshold	Show threshold trade-offs
Positive and negative sample counts	Make evidence assessable
Inference latency and peak memory	Demonstrate operational feasibility
Brier score and calibration assessment if feasible	Assess score reliability

If a partition contains only one class for a finding, its ROC AUC is undefined. Display that explicitly.

- Compare against a simple baseline.
- Inspect false positives and false negatives.
- Report subgroup performance only when metadata and sample sizes support it.
- Do not claim cross-site generalization from a single-site dataset.
- Do not equate 0.80 AUC with 80 percent accuracy.

Show model score until calibration has been demonstrated. An output of 0.85 is not automatically an 85 percent chance of injury.

7 Keep explanations optional and accurately named

Grad-CAM or slice attention can show what influenced a prediction. Neither establishes the precise location of an injury. Use Model attention or Influential slice. Avoid claims such as Tear detected here.

Explanations must correspond to the selected finding, study and checkpoint. Verify alignment with the displayed image. If explainability fails, retain the prediction and show Explanation unavailable. Never manufacture a convincing overlay.

8 Keep the software architecture lean

Layer	Recommended choice
Training and inference	Python and PyTorch
Data loading	Adapter for selected dataset; DICOM tooling only where needed
Backend	FastAPI
Frontend	React, TypeScript and Vite
Styling	Tailwind with accessible components
Viewer	Canvas with server-rendered slices initially
Persistence	Study manifest, checkpoint, metrics JSON and cached results
Packaging	One reproducible local or container setup

A separate Node gateway, Redis cluster and database are unnecessary for the bundled-study MVP.

API contract

Endpoint	Responsibility
GET /studies	Available studies
GET /studies/{id}	Series and supported metadata
GET /studies/{id}/series/{sid}/slices/{index}	Rendered slice
POST /predictions	Analyze a selected study
GET /studies/{id}/explanations/{finding}	Optional explanation
GET /metrics	Versioned evaluation snapshot
GET /model-card	Provenance and limitations
GET /healthz	Service health

Prediction responses include study ID, model and preprocessing versions, supported findings, scores, warnings, and result source: live or cached.

Measure synchronous inference before deciding it is sufficient. If it exceeds practical request limits, add a

minimal job and polling pattern. Keep uploaded or restricted data out of logs and public hosting unless explicitly permitted by the applicable terms.

9 Assign ownership by deliverable

Workstream	Accountable output
Data	Authorized dataset, audit, manifest and preprocessing
Modeling	Working checkpoint and reproducible training configuration
Labels and explainability	Label provenance and optional explanations
Evaluation	Split checks, metrics, model card and cached results
Backend	Documented API using the real inference pipeline
Frontend	Complete review workflow and demo

For a smaller team, merge related workstreams. Anas's proposed role is product and frontend plus integration, with explicit boundaries. This is a recommendation to agree with the team, not an assignment already accepted. It must not silently become responsibility for data acquisition, training, backend and everyone's unfinished tasks.

Freeze shared contracts before independent work starts: finding IDs, study IDs, tensor format, prediction schema, metrics schema, and slice and explanation coordinates.

Development fixtures should be conspicuously marked. They help integration, but must not quietly survive as final model results.

10 Define completion through evidence

- A teammate can reproduce setup and inference from the README.
- Selected studies pass through the same preprocessing used in training.
- The app displays real checkpoint outputs.
- Cached outputs identify their source and checkpoint.
- Missing or malformed inputs produce clear errors.
- Held-out evaluation includes counts and split methodology.
- Every displayed finding has a legitimate label source.
- Explanations are correctly aligned or omitted.
- Data and model assets are used within verified permissions.
- The demo shows a correct case, an error case and a limitation.

If the model performs poorly, report it. A useful failure analysis is stronger engineering evidence than inflated results.

Presentation narrative for the next step

Demonstrate the entire system: inspect a study, run inference, review a finding, reveal its dataset label, show evaluation, then explain a failure case. This provides substance beyond a polished dashboard and gives each team member a concrete contribution they can defend.

Keep a clearly labeled cached inference mode and a recorded demonstration as operational fallbacks. Cached results must never appear to be fresh inference. Preparing the slide deck is the next separate task.

Decisions still required

Decision	Evidence required
Dataset and usage permissions	Actual access and applicable training, hosting and demonstration terms
Supported finding list	Available labels and provenance
Evaluation partition	Patient grouping where available and adequate label counts
Serving mode	Measured latency and request limits
Named owners	Team agreement on deliverables and boundaries
Submission expectations	Organizer requirements for repository, demo, recording or other deliverables

Sources and visual provenance

[1] PRD RSNA Knee MRI Abnormality Detection (Hackathon).pdf. Supplied team draft dated 12 October 2026. Section 17 defines the standalone hackathon scope and supersedes earlier competition requirements.

[2] Kaggle. RSNA Knee Abnormality Detection, evaluation and dataset overview. Accessed 13 October 2026.
Competition evaluation
Dataset overview

[3] Stanford ML Group. MRNet dataset and competition. Published tasks and patient-grouped dataset split. Accessed 13 October 2026.
MRNet dataset and tasks

[4] RSNA. Knee MRI AI Challenge 2026. Official project context. Accessed 13 October 2026.
RSNA challenge overview

Figure 1 is an AI-generated UI concept created for this plan. Its scan, attention overlay and metadata are illustrative. Figure 2 is a deterministic diagram of the proposed architecture. Neither figure reports experimental results.